

CSRD Double Materiality Assessment Tool

Methodology & Framework Documentation

Built in accordance with ESRS 1 General Requirements under CSRD Directive 2022/2464/EU

1. Introduction

The Corporate Sustainability Reporting Directive (CSRD) represents the most significant expansion of sustainability reporting requirements in EU history. Effective from 2024 for large public-interest entities and phased in through 2028 for smaller companies, CSRD requires organisations to assess and disclose sustainability information across environmental, social, and governance dimensions — but with a distinctive methodology not found in predecessor frameworks like NFRD or voluntary standards like GRI.

At the core of CSRD compliance is the **double materiality assessment**: a structured process that evaluates sustainability topics from two directions simultaneously. This tool operationalises that process as an interactive, accessible web application, guiding users through each stage of the assessment in accordance with the European Sustainability Reporting Standards (ESRS).

This document explains the regulatory basis, conceptual framework, scoring methodology, sector benchmark logic, and limitations of the tool. It is intended to be read alongside the tool itself, and to demonstrate that the design decisions reflect genuine understanding of the underlying regulatory requirements.

2. Regulatory Context

2.1 The CSRD and Its Scope

The Corporate Sustainability Reporting Directive (Directive 2022/2464/EU) entered into force in January 2023, replacing the Non-Financial Reporting Directive (NFRD). It significantly expands both the scope of companies required to report and the depth of information they must disclose.

Mandatory reporting timelines are phased:

Timeline	Entity Type
FY2024 (reports in 2025)	Large public-interest entities already under NFRD (500+ employees)
FY2025 (reports in 2026)	Other large EU companies (250+ employees, or €40M+ turnover, or €20M+ assets)
FY2026 (reports in 2027)	Listed SMEs, small non-complex credit institutions, captive insurance undertakings
FY2028 (reports in 2029)	Non-EU companies with significant EU operations (€150M+ net turnover in EU)

The directive requires reporting in accordance with the European Sustainability Reporting Standards (ESRS), developed by the European Financial Reporting Advisory Group (EFRAG).

2.2 The European Sustainability Reporting Standards (ESRS)

ESRS are the technical standards specifying what companies must disclose under CSRD. They are organised into:

- **ESRS 1** — General Requirements (methodology, materiality, value chain)
- **ESRS 2** — General Disclosures (strategy, governance, stakeholder engagement)
- **ESRS E1–E5** — Environmental standards (climate, pollution, water, biodiversity, circular economy)
- **ESRS S1–S4** — Social standards (own workforce, value chain workers, communities, consumers)
- **ESRS G1** — Governance standards (business conduct)

ESRS 1 establishes the double materiality methodology that underpins this tool. All topic IDs and classifications in the tool correspond directly to these standards.

2.3 Relationship to Other Frameworks

CSRD double materiality is distinct from — but compatible with — other major sustainability frameworks:

Framework	Materiality Approach
CSRD / ESRS	Double materiality (impact + financial)
ISSB / IFRS S1–S2	Single materiality (financial only)
GRI Standards	Impact materiality (stakeholder-centric)
TCFD	Financial materiality (climate risk/opportunity)
EU Taxonomy	Alignment-based (not materiality)

Companies reporting under multiple frameworks can use this tool's impact and financial scores separately to satisfy GRI (impact dimension) and ISSB/TCFD (financial dimension) requirements alongside CSRD.

3. Double Materiality: Conceptual Framework

3.1 The Two Dimensions

ESRS 1 defines double materiality as the union of two distinct but related assessments:

Impact Materiality asks: *How does the company affect the world?*

A sustainability topic is impact-material if the company has actual or potential, positive or negative, short-, medium-, or long-term impacts on people or the environment — across its own operations and its upstream and downstream value chain. This dimension is rooted in stakeholder and societal concern, and aligns with the "inside-out" perspective on sustainability.

Financial Materiality asks: *How does the world affect the company?*

A sustainability topic is financially material if it generates or could generate financial risks or opportunities that affect — or could reasonably be expected to affect — the company's cash flows, access to finance, or cost of capital over the short, medium, or long term. This dimension aligns with investor-focused disclosure and the "outside-in" perspective.

3.2 The Union Principle

Under ESRS 1, a topic need only be material on **one** dimension to be considered material overall and require disclosure. This is the union principle — not the intersection. A topic that is highly impactful but carries no financial risk (e.g. a company's treatment of communities in a region where it has no financial exposure) is still material. Equally, a topic with significant financial risk but limited actual impact (e.g. a regulatory transition risk) is also material.

This tool reflects the union principle in its classification logic: a topic is flagged as Material if it meets or exceeds the materiality threshold on either the impact or financial dimension.

3.3 Value Chain Scope

ESRS 1 requires companies to consider their **full value chain** — not just direct operations — when assessing impact materiality. This includes:

- **Upstream:** raw material extraction, supplier manufacturing, logistics to the company
- **Own operations:** direct facilities, employees, processes
- **Downstream:** distribution, product use by consumers, end-of-life disposal

The tool's impact materiality scoring prompts users to consider this full value chain scope when rating each topic.

4. The ESRS Topic Structure

The tool covers the ten cross-cutting ESRS sustainability topics applicable to all sectors:

ID	Standard	Topic	Dimension
E1	ESRS E1	Climate Change	Environmental
E2	ESRS E2	Pollution	Environmental
E3	ESRS E3	Water & Marine Resources	Environmental
E4	ESRS E4	Biodiversity & Ecosystems	Environmental
E5	ESRS E5	Resource Use & Circular Economy	Environmental
S1	ESRS S1	Own Workforce	Social
S2	ESRS S2	Workers in the Value Chain	Social
S3	ESRS S3	Affected Communities	Social
S4	ESRS S4	Consumers and End-users	Social
G1	ESRS G1	Business Conduct	Governance

In addition to these cross-cutting topics, EFRAG is developing sector-specific ESRS standards for 40 sectors. The sector benchmark pre-populations in this tool are informed by EFRAG's sector-specific draft guidance, anticipating which topics are most likely to be material by industry type.

5. Scoring Methodology

5.1 Rating Scale and Inputs

Each topic is assessed on two sub-dimensions per materiality type, producing four scores per topic:

Input	Description	Scale
Impact Severity	How severe are the company's impacts on people/environment?	1 (Negligible) – 5 (Severe)
Impact Likelihood	How likely or certain are these impacts to occur?	1 (Unlikely) – 5 (Likely/Ongoing)
Financial Magnitude	How large could the financial effect be on the company?	1 (Negligible) – 5 (Severe)
Financial Likelihood	How likely is this financial effect to materialise?	1 (Unlikely) – 5 (Likely/Certain)

The 1–5 scale is intentional. ESRS 1 does not prescribe a specific numeric methodology, but recommends assessing impacts based on their scale, scope, and irremediability (for negative impacts) or scale and likelihood (for positive impacts). The two-factor approach of severity × likelihood is a widely adopted operationalisation of these criteria, consistent with EFRAG's technical guidance and practitioner frameworks used by Big Four sustainability advisory practices.

5.2 Score Computation

For each topic, a composite score is calculated for each dimension:

$$\text{Impact Score} = (\text{Impact Severity} + \text{Impact Likelihood}) / 2$$

$$\text{Financial Score} = (\text{Financial Magnitude} + \text{Financial Likelihood}) / 2$$

Both scores are rounded to one decimal place. This averaging approach weights severity and likelihood equally, reflecting the ESRS 1 principle that both the scale of an impact and the certainty of its occurrence are relevant to materiality.

5.3 Classification Tiers

Tier	Condition	Implication
Material	Impact Score $\geq$ 3.5 OR Financial Score $\geq$ 3.5	Must be disclosed in CSRD report under relevant ESRS
Potentially Material	Impact Score $\geq$ 2.5 OR Financial Score $\geq$ 2.5	Requires further assessment; monitor closely
Monitor	Both scores $<$ 2.5	Not currently material; review annually

Default thresholds of 3.5 (material) and 2.5 (potential) are aligned with common practitioner practice, where a score above the midpoint of a 1–5 scale in either sub-dimension signals meaningful concern. These thresholds are adjustable in the tool, recognising that different industries, assurance providers, and company risk appetites may warrant different cut-offs.

5.4 The Double Materiality Matrix

The visual output of the assessment is a scatter plot placing each topic at coordinates (Financial Score, Impact Score). This "double materiality matrix" is the standard visualisation format used by CSRD practitioners and referenced in EFRAG's illustrative guidance. Colour-coding by tier (red = material, amber = potential, grey = monitor) allows at-a-glance identification of priority topics.

6. Stakeholder Engagement

ESRS 1 explicitly requires that companies involve stakeholders in the materiality assessment process. Specifically, companies must:

- Identify affected stakeholders and stakeholders with an interest in the company's sustainability performance
- Consult with these groups or their legitimate representatives
- Document the process, findings, and how stakeholder input influenced materiality conclusions

The tool's stakeholder consultation section captures the minimum information required to demonstrate a credible process: groups consulted, consultation period, methodology used, and key findings. This information appears in the generated PDF report, providing an audit trail for external assurance providers.

Stakeholder groups covered in the tool reflect the categories commonly identified in ESRS 2 and EFRAG's supplementary guidance:

- Employees & Trade Unions (own workforce representatives)
- Investors & Lenders (capital providers)
- Customers & End-users (downstream value chain)
- Suppliers & Partners (upstream value chain)
- Local Communities (affected communities)
- NGOs & Civil Society (civil society representatives)
- Regulators & Government (public authorities)
- Industry Associations (sector bodies)
- Academic & Research (expert input)

7. Sector Benchmark Pre-populations

To assist users in establishing a defensible starting point for assessment, the tool pre-populates scores when a sector is selected. These benchmarks are informed by:

- EFRAG's sector-specific ESRS exposure drafts and technical explanations
- The EU Taxonomy sector classifications and their identified environmental objectives
- Published double materiality assessments from companies in each sector (where publicly available)
- Academic literature on sector-level ESG materiality

The pre-populated values represent topics that are **typically** material for companies in each sector, based on their common business activities and value chains. They are explicitly labelled as starting points and should be reviewed and adjusted to reflect the specific circumstances of the company being assessed.

Example benchmark rationale for Financial Services: G1 (Business Conduct) and S4 (Consumers & End-users) are pre-populated at 4 because financial institutions face significant regulatory and reputational exposure to anti-money laundering, mis-selling, and data privacy failures. E1 (Climate Change) is pre-populated at 3 reflecting growing financed emissions requirements under the ECB's supervisory expectations and EU taxonomy alignment obligations.

8. Year-over-Year Comparison

The tool supports loading a prior year's saved assessment to generate a comparative analysis. This feature reflects the CSRD requirement for companies to report material topics consistently across reporting periods and to explain changes in their materiality conclusions.

The comparison view shows:

- Movement of each topic on the double materiality matrix (visualised with directional arrows)
- Delta scores for both impact and financial dimensions
- Tier changes (e.g. a topic moving from Potentially Material to Material)

This longitudinal view is particularly useful for companies in their second or third year of CSRD reporting, where demonstrating an evolving understanding of material topics, and the actions taken in response, is expected by assurance providers and investors.

9. Limitations and Caveats

This tool is designed for educational use and as a structured starting point for a real materiality assessment. Users should be aware of the following limitations:

Quantitative simplification. ESRS 1 acknowledges that materiality assessment involves both quantitative and qualitative judgement. The 1–5 numeric scoring approach simplifies what is often a nuanced, evidence-based process involving site-level data, legal analysis, and expert consultation. Scores should be understood as structured approximations, not precise measurements.

Sector benchmarks are generalisations. Pre-populated scores reflect typical sector patterns but cannot account for company-specific business models, geographies, or supply chain configurations. A financial services company with significant data centre operations, for example, may have higher E1 and E2 exposure than the sector average.

Value chain data limitations. Accurate impact materiality assessment — particularly for S2 (Value Chain Workers) and E4 (Biodiversity) — typically requires supplier-level data that many companies do not yet have. The tool cannot substitute for this data collection.

Not a substitute for professional assessment. CSRD reports used for official regulatory purposes must be prepared with appropriate professional expertise and subject to external assurance (limited assurance from 2024, reasonable assurance expected from 2028). This tool does not constitute professional sustainability reporting advice.

Dynamic regulatory environment. ESRS standards and EFRAG sector-specific guidance continue to evolve. The tool reflects the standards as understood at time of publication (February 2026) and should be reviewed against any subsequent regulatory updates.

10. References & Further Reading

Regulatory Sources

- European Parliament & Council. (2022). *Corporate Sustainability Reporting Directive* (Directive 2022/2464/EU). Official Journal of the European Union.
- European Commission. (2023). *Commission Delegated Regulation (EU) 2023/2772 — European Sustainability Reporting Standards*. Official Journal of the European Union.
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Guidance & Practitioner Resources

- EFRAG. (2024). *Materiality Assessment Implementation Guidance*. European Financial Reporting Advisory Group.
- GRI & EFRAG. (2023). *Interoperability Guidance: GRI Standards and ESRS*. Global Reporting Initiative.
- World Economic Forum. (2023). *Embracing the CSRD: A Practitioner's Guide to Double Materiality*.

Academic & Conceptual Background

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- Drempetic, S., Klein, C., & Zwergel, B. (2020). The influence of firm size on the ESG score. *Journal of Business Ethics*, 167(2), 333–360.
- Fasan, M., & Mio, C. (2017). Fostering stakeholder engagement: The role of materiality disclosure in integrated and sustainability reporting. *Business Strategy and the Environment*, 26(3), 288–305.

Related Tools & Standards

- EU Taxonomy Regulation (2020/852/EU) — for alignment assessment alongside materiality
- ISSB IFRS S1 & S2 — for financial materiality under the global baseline standard
- GRI Universal Standards (2021) — for impact materiality and stakeholder engagement guidance

This methodology document was prepared as a companion to the CSRD Double Materiality Assessment Tool. For questions or feedback, contact [your name] at [your email] or visit [your portfolio URL].